

Instant CS Fundamentals Test

Time: 120 minutes | Total Marks: 70

Section A – Problem Solving (40 pts) → (Choose Two Problems To Solve)

Problem 1 – Expression Evaluator (20 pts)

Problem Statement :

You are given a string representing a mathematical expression containing:

- digits (0-9)
- '+' and '-' operators
- parentheses '(' and ')'

Evaluate the expression and print the result.

You must use a stack to solve the problem.

Example:

Input:

(1+(2-3)+4)

Output: 4

Constraints:

- Expression length $\leq 10^5$
- No spaces in the input
- The expression is always valid

Problem 2 – Customer Service Simulation (20 pts)

Problem Statement :

A service center handles customer requests using a queue.

You are given Q operations:

- ENQUEUE x → add customer x to the queue
- SERVE → remove the front customer
- FRONT → print the customer at the front (or print "Empty" if none)

After processing all operations,

print the result of every FRONT command.

Example:

Input:

6

ENQUEUE 5

ENQUEUE 10

FRONT

SERVE

FRONT

SERVE

Output: 5 10

Problem 3 – Student Ranking System (20 pts)

Problem Statement :

You are given data about N students.

Each student has:

- Name (string without spaces)
- Score (integer)

Store students using `vector<pair<string, int>>`.

Sort students by:

1. Higher score first
2. If scores are equal → lexicographically smaller name first

Then print the sorted list.

Example:

Input:

3

Ali 90

Sara 95

Omar 90

Output:

Sara 95

Ali 90

Omar 90

Constraints:

- $1 \leq N \leq 10^5$
- Name length ≤ 20
- Score: 0 to 100

Section B – Multiple Choice Questions (12 pts)

1. Which STL container does NOT support iterators?

A) vector B) deque C) stack D) none of the above

2. Which operation is NOT directly supported by a stack?

A) push B) pop C) top D) random access

3. What happens when you access an out-of-range index in a vector?

- A) Compilation error
B) Runtime error or undefined behavior
C) Automatically resizes
D) Returns 0

4. Which STL container allows insertion/removal from both ends efficiently?

A) vector B) deque C) stack D) pair

5. Which data structure follows the LIFO principle?

A) Queue B) Stack C) Deque D) Vector

6. What is the average time complexity of accessing a vector element by index?

A) $O(1)$ B) $O(\log n)$ C) $O(n)$ D) $O(n^2)$

Section D – Short Answer (18 pts)

1. What is the advantage of using STL containers instead of raw arrays?
2. Explain the difference between vector and deque.
3. When would you prefer using a stack over a queue?
4. What is time complexity? Why is it important?
5. What happens internally when a vector runs out of capacity?
6. What is the difference between pass by value and pass by reference?